

# Deep Learning Research Engineer

## Technical Assignment

*LLM Optimization, Quantization & NPU Model Porting*

| Field               | Details                                        |
|---------------------|------------------------------------------------|
| Role                | Deep Learning Research Engineer                |
| Assignment Version  | v1.0 - Gemma 4 E2B Series                      |
| Estimated Duration  | 8–12 hours                                     |
| Submission Deadline | [Insert Date — typically 5 business days]      |
| Evaluation Areas    | Quantization · Model Conversion · Benchmarking |

### Overview & Instructions

This assignment evaluates your practical skills in LLM optimization, model conversion, and inference benchmarking using google/gemma-4-E2B. Gemma 4 is a modern decoder-only model with long-context support, KV-cache based generation, and practical deployment paths that still present non-trivial real-world challenges.

There are three core challenges plus an open-ended section. Use any open-source Python libraries you like, and use google/gemma-4-E2B as the primary model. If your hardware is limited, you may work on a reduced setup for benchmarking or use smaller representative sequence lengths, but keep the model choice fixed and document your decisions and trade-offs throughout.

Submit a single ZIP containing: all source code, a written report (PDF or Markdown), benchmark results (CSV or JSON), and a README with setup instructions and hardware details.

### Background: Gemma 4 E2B Architecture

Gemma 4 E2B is a decoder-only generative model from Google DeepMind. In practice, this assignment should focus on the text-generation path: token embedding, decoder blocks, attention, MLP layers, and KV-cache based autoregressive decoding. The main optimization

challenges are long-context efficiency, quantization trade-offs across transformer components, and maintaining numerically stable generation behavior after export or edge conversion.

You are expected to research the architecture as needed. Primary model on HuggingFace: google/gemma-4-E2B. Unless clearly justified otherwise, keep the assignment centered on this exact checkpoint rather than switching to another Gemma variant.

## Challenge 1: LLM Quantization & Optimization

*Estimated Time: 3–4 hours | Weight: 35%*

Quantize google/gemma-4-E2B using at least two strategies (e.g. INT8 PTQ, INT4, or mixed-precision). For each, report model size, inference latency, and perplexity or another clearly justified text-quality metric vs. the fp16 or bf16 baseline. Then perform a layer-wise sensitivity analysis - quantize one layer type at a time and explain why certain components (for example attention projections, MLP blocks, embeddings, or lm\_head) are more sensitive. Finally, apply at least one inference optimization (e.g. torch.compile, KV-cache optimization, or speculative decoding) and measure its impact.

| Criterion                              | Weight |
|----------------------------------------|--------|
| Quantization correctness & coverage    | 35%    |
| Sensitivity analysis depth & reasoning | 35%    |
| Optimization impact & benchmarking     | 30%    |

## Challenge 2: Model Conversion (ONNX / Edge Format)

*Estimated Time: 3–4 hours | Weight: 35%*

Export google/gemma-4-E2B to ONNX with dynamic axes and correct KV-cache input/output handling, then convert to one edge/mobile format of your choice - Apple MLX, LiteRT-LM, or QNN (Qualcomm AI Hub). Validate outputs against the PyTorch reference (within 1e-3 tolerance for logits on a representative text-only test case).

Then write a short Conversion Notes section (200–400 words) describing the most significant obstacle you hit, what you tried, and what a full production deployment would still need. Honest, detailed failure analysis is valued over a polished-but-vague success story.

| Criterion                                                             | Weight |
|-----------------------------------------------------------------------|--------|
| ONNX export correctness (dynamic axes, KV-cache handling, validation) | 35%    |
| Edge format conversion & output validation                            | 35%    |

| Criterion                        | Weight |
|----------------------------------|--------|
| Conversion Notes depth & honesty | 30%    |

## Challenge 3: Inference Throughput Benchmarking

*Estimated Time: 2–3 hours | Weight: 30%*

Build a benchmarking harness that measures Time to First Token (TTFT), Tokens Per Second (TPS), and peak memory across a range of precision, batch size, and sequence length configurations. You don't need to run every combination — choose a representative subset of at least 12 and justify your selection. Each configuration should run at least 5 times with proper warmup; report mean  $\pm$  std deviation.

Save results as a CSV: model, precision, batch\_size, input\_seq\_len, gen\_len, ttft\_ms\_mean, ttft\_ms\_std, tps\_mean, tps\_std, peak\_mem\_mb, hardware, notes

Close with a short written analysis (300–500 words): which configuration best suits a real-time chatbot, what the memory bottleneck looks like at large batch sizes or long context lengths, and how Gemma 4 E2B's decoder-only attention design affects throughput scaling as prefill and decode loads change.

| Criterion                                         | Weight |
|---------------------------------------------------|--------|
| Benchmark harness correctness & statistical rigor | 40%    |
| Coverage & justification of test configurations   | 25%    |
| Depth of written analysis                         | 35%    |

## Bonus: Bring Your Own Idea

*Optional — no right answer. This is where we want to see you.*

Outside of the three core challenges, show us something that excites you. This could be anything that extends, improves, or adds something unexpected to the work above. A few sparks if you need them are listed below, including an optional multimodal extension that can be used as bonus work rather than a core requirement:

A multimodal extension using image + text inputs, such as visual question answering, captioning, or a short note on quantization / conversion trade-offs for a vision-language pipeline built around Gemma or a compatible multimodal setup

- An NPU deployment demo (QNN, Apple MLX, or LiteRT-LM) with power profiling
- A creative visualization of quantization error or layer sensitivity
- A novel caching or batching strategy specific to transformer KV-cache behavior

- A side-by-side comparison of Gemma 4 E2B vs. another open LLM on the same benchmarks
- Anything else that showcases how you think about ML systems

There are no constraints here. The only thing we ask is that you briefly explain what you built and why you found it interesting.

## Overall Evaluation Rubric

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| Challenge                                | Weight     |
|------------------------------------------|------------|
| Challenge 1: Quantization & Optimization | 35%        |
| Challenge 2: Model Conversion            | 35%        |
| Challenge 3: Benchmarking                | 30%        |
| Bonus: Bring Your Own Idea               | +up to 10% |

*We care as much about how you reason and communicate as about the results themselves. A submission that explains what it tried, what failed, and why is more valuable than one that silently skips the hard parts.*

## Submission Checklist

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| Item                                      | Required?   |
|-------------------------------------------|-------------|
| Source code + scripts                     | Yes         |
| README.md (setup, hardware, known issues) | Yes         |
| Written report (PDF or Markdown)          | Yes         |
| Benchmark CSV                             | Yes         |
| requirements.txt                          | Yes         |
| ONNX / edge format model                  | Recommended |

***Good luck! We look forward to reviewing your submission.***